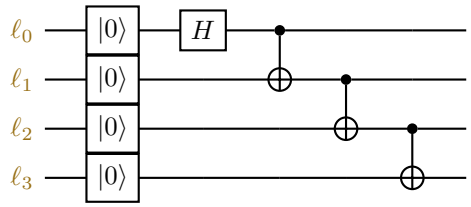
Shared gadget: reset, cat preparation, and verification

Read every circuit left to right. A box $|0\rangle$ means reset, discarding the previous state; it is not a projection with postselection. A meter measures Z. The sequence H then meter measures X. Single wires are quantum; double wires carry classical bits.

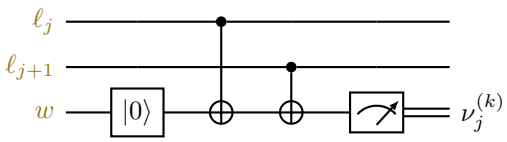
$$\mathcal{R}_0(\rho) = |0\rangle\langle 0| \operatorname{Tr}\rho, \qquad b = 0 \iff \text{reported sign } +1.$$

Prepare a raw cat: the four-leaf example



For m leaves, extend the chain through ℓ_{m-1}: H_{ℓ_0} followed by CNOTs $\ell_j \to \ell_{j+1}$, $j = 0, \ldots, m-2$.

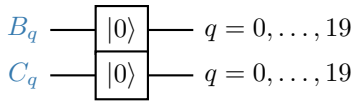
One adjacent-ZZ verification, with its own reset



$$\text{For } k = 1, 2, 3: \quad j = 0, 1, \ldots, m-2; \qquad \text{require every } \nu_j^{(k)} = 0.$$

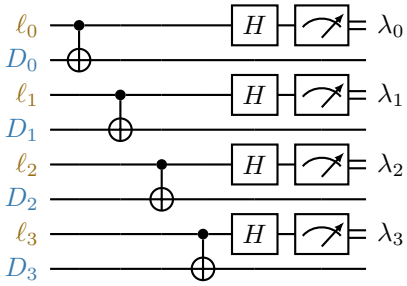
Reset w before every pair. After verification, apply no further leaf-to-leaf gates. Couple each leaf to one target port only, then measure all leaves in X. The check bit is $y = \bigoplus_j \lambda_j$. For a positive check require $y = 0$, not every $\lambda_j = 0$.

1. Prepare the encoded B and C inputs

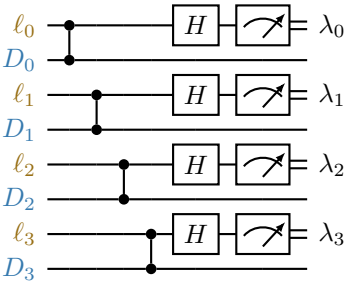


The drawing represents an independent reset of every physical data qubit. For each check below, prepare one fresh verified cat as on page 1. The leaves in each panel are one joint cat, not independent plus states.

$X_{D_0}X_{D_1}X_{D_2}X_{D_3}$



$Z_{D_0}Z_{D_1}Z_{D_2}Z_{D_3}$



Full target lists and repetitions

For $s = 0, \ldots, 8$, replace $(0, 1, 2, 3)$ by the ordered support R_s from page 13. An upper check has eight leaves. Logical checks have six.

$$\mathcal{F}_B = (S^X_{B,0:8},\ S^Z_{B,0:8},\ \overline{Z}^B_0, \overline{Z}^B_1),$$
$$\mathcal{F}_C = (S^X_{C,0:8},\ S^Z_{C,0:8},\ \overline{X}^C_0, \overline{Z}^C_1).$$

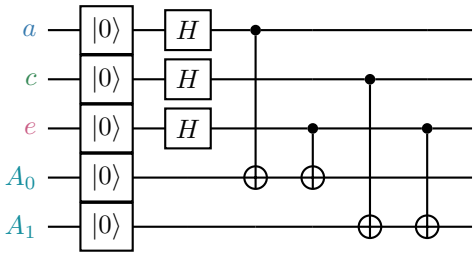
The notation 0:8 means all nine checks in increasing s order. Execute three complete positive rounds on B, then three on C. Require all check bits zero, in addition to all cat-verification bits.

$$|\psi\rangle = |\overline{00}\rangle_B |\overline{+0}\rangle_C.$$

2. Prepare the graph resource and verify its edge checks

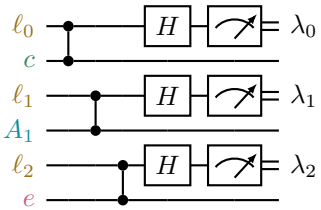
Reset a, b, c, d, e and $A_0, \ldots, A_{19}$ to zero. Apply H to each vertex. For every edge $q : u_q - v_q$, apply CNOT$(u_q \to A_q)$ then CNOT$(v_q \to A_q)$, in increasing q.

The gates for edges 0 and 1 (other edges continue the same wires)



Endpoints	Labels	Endpoints	Labels
ab	7,15	bc	8,19
ac	11,12	bd	14,16
ad	3,6	be	4,9
ae	0,10	cd	2,13
ce	1,18	de	5,17

One edge-check measurement: $K_1 = Z_c Z_{A_1} Z_e$



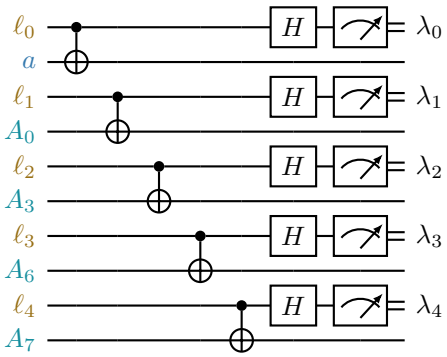
One three-leaf verified cat is shared by these three ports. Set $y = \lambda_0 \oplus \lambda_1 \oplus \lambda_2$ and require zero. For edge $q : u - v$, use targets (u, A_q, v). The vertex checks on the next page complete resource verification.

2 continued. Verify a resource vertex check

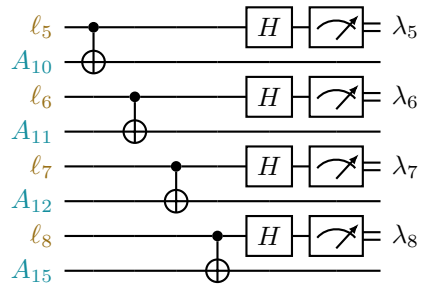
$$K_a = X_a X_{A_0} X_{A_3} X_{A_6} X_{A_7} X_{A_{10}} X_{A_{11}} X_{A_{12}} X_{A_{15}}.$$

Prepare **one nine-leaf cat**. The two panels show its first five and last four target ports. They do not prepare or measure two separate cats.

Leaves 0–4



Leaves 5–8



$$y(K_a) = \lambda_0 \oplus \cdots \oplus \lambda_8; \qquad \text{require } y(K_a) = 0.$$

Use the same circuit for each vertex v, replacing a by v and $E_a = R_5$ by E_v:

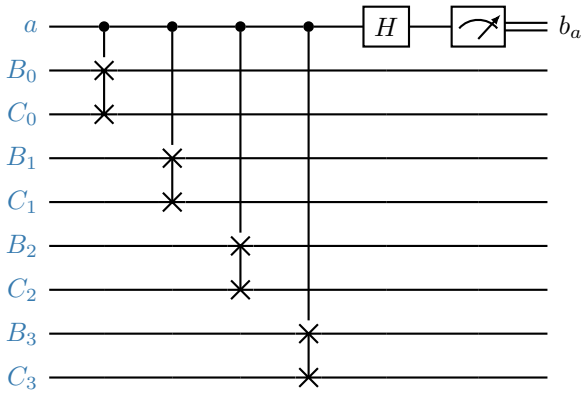
$$K_v = X_v \prod_{q \in E_v} X_{A_q}, \qquad (E_a, E_b, E_c, E_d, E_e) = (R_5, R_6, R_7, R_8, R_9).$$

The complete resource-verification schedule

Measure $(K_a, K_b, K_c, K_d, K_e, K_0, \ldots, K_{19})$ in that order, with a fresh verified cat for every check. Execute three positive rounds. These are 3×25 check results, all required zero. The resource is $2^{-5/2} \sum_t |t\rangle_V |\delta t\rangle_A$.

3. Enrich the symmetry and measure the temporary vertices

The a vertex coherently controls four physical SWAPs. Each column with one dot and two crosses is a three-qubit Fredkin.



Vertex	Data-pair labels J_v	Quantum action before its X readout
a	0–3	$\mathrm{C}_a(T_a)$
b	4–7	$\mathrm{C}_b(T_b)$
c	8–11	$\mathrm{C}_c(T_c)$
d	12–15	$\mathrm{C}_d(T_d)$
e	16–19	$\mathrm{C}_e(T_e)$

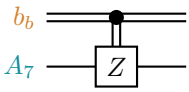
Apply **all twenty Fredkins first**, increasing q. Then read vertices a,b,c,d,e in X. The meter in the panel shows the a readout; defer it until all twenty enrichment Fredkins have finished. The gauge qubits are idle during enrichment.

$$r_{\mathrm{bit}} = b_a \oplus b_b \oplus b_c \oplus b_d \oplus b_e; \qquad \text{require } r_{\mathrm{bit}} = 0.$$

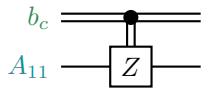
4. Use the vertex reports to fix the gauge signs

The upper wires below are the already measured bits, not surviving vertex qubits. A double-wire control applies Z when its bit is one.

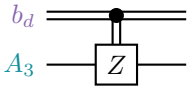
b_b **controls** Z_{A_7}



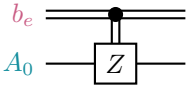
b_c **controls** $Z_{A_{11}}$



b_d **controls** Z_{A_3}



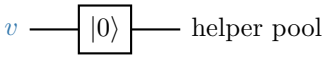
b_e **controls** Z_{A_0}



$$F_b = Z_{A_7}^{b_b} Z_{A_{11}}^{b_c} Z_{A_3}^{b_d} Z_{A_0}^{b_e}.$$

Apply these four possible gates in the displayed order. There is no root-a correction. The already imposed even-parity condition fixes its remaining Gauss sign.

Release the five vertex qubits



After their readout, all five vertices may be reset for helper reuse. From this point on, a–e name graph locations and data groups; they are not five live gauge-check ancillas.

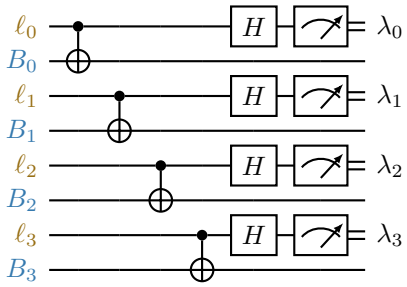
$$W = \frac{1}{4} \sum_{t:t_a=0} |\delta t\rangle_A V_t, \qquad |\text{gauged}\rangle \propto W \frac{I+U}{2} |\psi\rangle.$$

5a. While gauged: measure the paired code checks

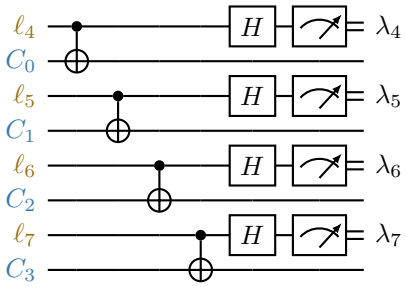
$$N_0^X = X_{B_0}X_{B_1}X_{B_2}X_{B_3}X_{C_0}X_{C_1}X_{C_2}X_{C_3}.$$

Prepare **one eight-leaf cat** across both panels. These couplings produce one eight-leaf parity, not two four-leaf results.

Leaves 0–3: B targets



Leaves 4–7: C targets



$$y(N_0^X) = \bigoplus_{j=0}^{7} \lambda_j; \qquad \text{require } y(N_0^X) = 0.$$

The complete family

For $R_s = (q_0 < \cdots < q_{h-1})$, use $2h$ leaves, with ports

$$\ell_j \to B_{q_j}, \qquad \ell_{h+j} \to C_{q_j}, \qquad j = 0, \ldots, h-1.$$

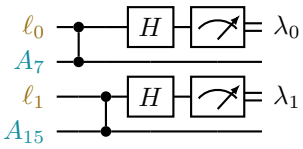
For N_s^X use CNOT; for N_s^Z replace each CNOT target by a CZ dot. Measure $N_0^X, \ldots, N_8^X$, then $N_0^Z, \ldots, N_8^Z$. The largest such cat has sixteen leaves.

Do not measure $S_{B,s}$ and $S_{C,s}$ separately while gauged. The required observable is their product. Cycle and Gauss checks follow, then the entire 39-check round is repeated, three complete rounds total.

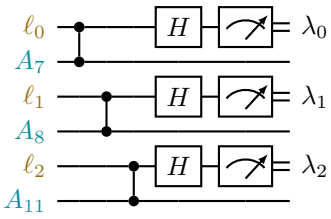
5b. While gauged: measure the sixteen cycle checks

Use one fresh verified cat for each cycle. All target gates are CZs. The measurement acts only on gauge qubits.

$L_{(7,15)} = Z_{A_7} Z_{A_{15}}$



$L_{(7,8,11)} = Z_{A_7} Z_{A_8} Z_{A_{11}}$



All sixteen supports, in measurement order

$$(7,15),\ (11,12),\ (3,6),\ (0,10),\ (8,19),$$
$$(14,16),\ (4,9),\ (2,13),\ (1,18),\ (5,17),$$
$$(7,8,11),\ (7,14,3),\ (7,4,0),$$
$$(11,2,3),\ (11,1,0),\ (3,5,0).$$

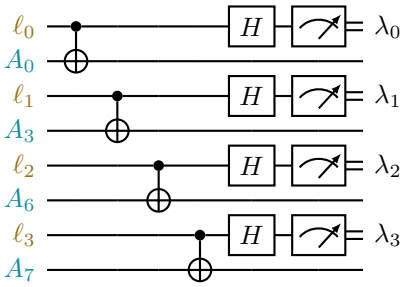
Within a check, couple ports in increasing edge-label order, then measure the leaves. Require the single parity $y(L_\eta) = \bigoplus_j \lambda_j = 0$. The six triangles together with the ten parallel-edge pairs form a cycle basis. After these sixteen checks, measure the five Gauss checks.

5c. While gauged: the full local Gauss-check circuit

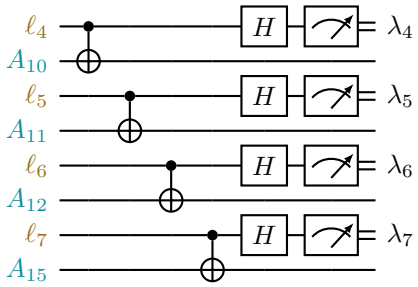
$$G_a = (X_{A_0}X_{A_3}X_{A_6}X_{A_7}X_{A_{10}}X_{A_{11}}X_{A_{12}}X_{A_{15}})T_a.$$

Prepare **one twelve-leaf cat** before all four panels. Eight ports are CNOTs to gauge qubits; four are Fredkins to data pairs.

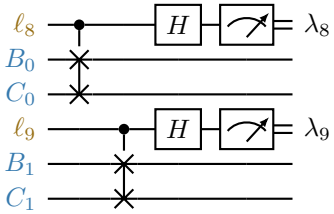
Gauge ports 0–3



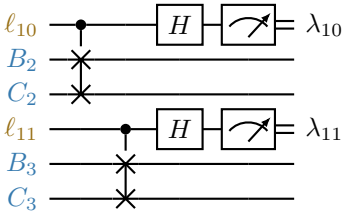
Gauge ports 4–7



Data-pair ports 8–9



Data-pair ports 10–11



$$y(G_a) = \bigoplus_{j=0}^{11} \lambda_j; \qquad \text{require } y(G_a) = 0.$$

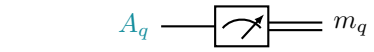
The panels are port fragments, not four check measurements. Execute all twelve controlled factors before the leaf readout. For v=b,c,d,e replace (E_a, J_a) by (E_v, J_v), retaining the same twelve-port pattern.

$$G_v = X_{A,E_v}T_v, \qquad G_aG_bG_cG_dG_e = U.$$

One full round is 18 N checks, 16 L checks, then 5 G checks. Repeat that entire round three times, using fresh cats throughout.

6. Ungauge: read all edges, decode, and correct

Measure every A_q in Z, increasing $q = 0, \ldots, 19$. Collect all twenty reported bits before choosing any SWAP correction.

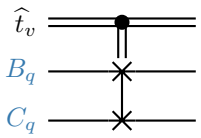




The double-wire box is classical computation. Enumerate all sixteen cut words $w = \delta t$, with $t_a = 0$. Choose the word $\widehat{w}$ within Hamming distance two of m, or reject if none exists.

$$\widehat{t}_a = 0, \quad \widehat{t}_b = \widehat{w}_7, \quad \widehat{t}_c = \widehat{w}_{11}, \quad \widehat{t}_d = \widehat{w}_3, \quad \widehat{t}_e = \widehat{w}_0.$$

A classical bit controls each ordinary data SWAP



For each $q \in J_v$, apply SWAP(B_q, C_q) when $\widehat{t}_v = 1$, omitting it otherwise. Execute $q = 0, \ldots, 19$ in order. These are not coherent Fredkins: their controls are classical bits. Discard/reset A.

Decode all twenty reports first; do not use the four raw tree bits.

7–8. Filter the original code, then recover

Reuse the four-body Pauli circuit on page 2, extended to eight targets for the upper checks. A fresh verified cat is required each time.

$$\mathcal{O} = (S^X_{B,0:8}, S^Z_{B,0:8}, S^X_{C,0:8}, S^Z_{C,0:8}).$$

7. Terminal positive filter

Measure the complete 36-check family $\mathcal{O}$ in three rounds. Require all 3×36 check bits zero. A nonzero syndrome rejects.

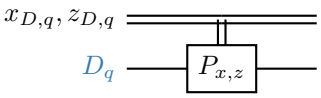
8. Subsequent syndrome recovery

Measure $\mathcal{O}$ repeatedly, retaining either check sign. Require cat-verification bits zero. Apply no correction between rounds.

$$\tau = \min\{r \in \{3, \ldots, 9\} : \mathbf{s}^{(r)} = \mathbf{s}^{(r-1)} = \mathbf{s}^{(r-2)}\}.$$

Reject if there is no identical triple by round nine. For each block D, choose a minimum-weight Pauli R_D, of weight at most two, matching its eighteen stopping-syndrome bits; reject if none exists. Use the fixed lookup in `c4_helix_intrinsic_decoder.py`.

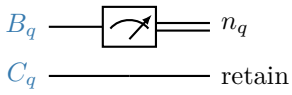
The physical correction: one-qubit gates only

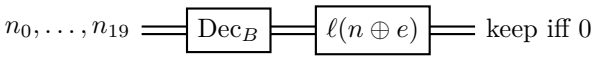


The classical pair selects $P_{0,0} = I$, $P_{1,0} = X$, $P_{0,1} = Z$, or $P_{1,1} = Y$. Apply the selected single-qubit Pauli, visiting physical labels in order, B then C; omit identity factors. For Y use one Y gate. The gate acts on one data qubit; the syndrome lookup is classical.

9. Decode the final B readout and retain magic in C

For $q = 0, \ldots, 19$, measure B in Z. C remains a quantum output. The horizontal C wire also represents its counted waiting idles.





$$
\begin{aligned}
h_s(n) &= \bigoplus_{q \in R_s} n_q, \qquad s = 0, \ldots, 8, \\
&\text{choose minimum-weight } e, \quad |e| \le 2, \quad h(e) = h(n), \\
n' &= n \oplus e, \\
\ell(n') &= n'_0 \oplus n'_3 \oplus n'_6 \oplus n'_7 \oplus n'_{18} \oplus n'_{19}.
\end{aligned}
$$

Reject if the correction e does not exist. Fix readout ties by lexicographic order of the flipped-position lists. Keep only $\ell(n') = 0$. Correction of n is classical; no postmeasurement X gates are applied to B. Discard B and retain the twenty-qubit C block.

Ideal logical output

$$
\frac{2|00\rangle + |01\rangle + |10\rangle}{\sqrt{6}} \xrightarrow{\overline{Z}_0^B = +1} |0\rangle_B \frac{2|0\rangle + |1\rangle}{\sqrt{5}}_C .
$$

Both second logical qubits are spectators in zero. The retained output is $|\overline{M0}\rangle_C$, up to the physical correctable errors allowed by the linked two-fault analysis.

Reference: supports, order, and resources

Each support defines $S^X_{D,s} = \prod_{q \in R_s} X_{D_q}$ and $S^Z_{D,s} = \prod_{q \in R_s} Z_{D_q}$. Measure only $s = 0, \ldots, 8$ as independent code checks; use row 9 for graph incidence.

s	R_s
0	0, 1, 2, 3
1	4, 5, 6, 7
2	8, 9, 10, 11
3	12, 13, 14, 15
4	16, 17, 18, 19
5	0, 3, 6, 7, 10, 11, 12, 15
6	4, 7, 8, 9, 14, 15, 16, 19
7	1, 2, 8, 11, 12, 13, 18, 19
8	2, 3, 5, 6, 13, 14, 16, 17
9	0, 1, 4, 5, 9, 10, 17, 18

$$\overline{X}_0 = X_{\{2,3,8,11,12,15\}},$$
$$\overline{Z}_0 = Z_{\{0,3,6,7,18,19\}}, \qquad \overline{Z}_1 = Z_{\{2,3,8,11,12,15\}}.$$

One attempt, in execution order

1. Prepare B, then C: three positive 20-check rounds each.
2. Prepare graph; three positive 25-check resource rounds.
3. Twenty enrichment Fredkins; measure the five vertices in X.
4. Require even vertex parity; apply the four possible gauge-Z gates.
5. Three positive gauged rounds: $18N$, $16L$, $5G$, in that order.
6. Gauge-Z readout; decode; apply ordinary data SWAPs.
7. Three positive 36-check original-code rounds.
8. Recovery: first identical syndrome triple, at most nine rounds.
9. Complete B-Z readout; decode; keep logical zero and retain C.

Allocation: 40 data + 20 gauge + 17 reusable helpers = 77 qubits. Five helpers initially serve as vertices. The largest cat has 16 leaves plus one verifier. **Fredkins:** $20 + 3 \times 5 \times 4 = 80$.

All gates and measurements are scheduled serially; paired panels are fragments of a circuit, not additional parallel hardware. Unused live qubits have counted idles. Any required failed test rejects the whole attempt. The global fault allowance is two faulty locations total, including helpers, reports, recovery, and readout, as specified in the companion analysis. These drawings do not change that protocol or prove a new fault-tolerance result.